

Michael Ma 马成军

Senior Backend Engineer / Tech Lead · High-Concurrency Transaction Systems & AI Engineering

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 LinkedIn  Tech Blog

Profile

13 years of backend engineering focused on **high-concurrency transaction systems** and **distributed architecture**. Having led incident response on a large-scale outage and maintained core transaction systems for years, I know what unreliable output costs in production — so when I extended into **AI engineering** this past year, the first thing I built was a convergence layer and cost controls, not another generator.

13 yrs Backend engineering · OTA / e-commerce / MarTech	¥10B+ GMV within 5 months, Apple China reseller platform	25s → 2s Critical API p99 after major-outage response
200% YoY growth, KKday affiliate business	Review −50% Maestro converging 4 AI review bots, 300+ PRs	20+ Daily users of self-built AI platforms, 7 teams

- AI engineering:** built a convergence layer for multiple AI review bots (Maestro) and an AI-assisted root-cause monitoring platform (VM Health Center), solving the practical problem of duplicated, unreliable, unconverged AI output
- Transaction-systems depth:** top contributor and owner of KKday's affiliate marketing platform (**200%** YoY growth), and owner of the B2C core transaction flow

AI Engineering

- Maestro** (TypeScript CLI) — a **convergence layer for AI review bots**: orchestrates **4** bots (CodeRabbit, Copilot, Codex, Gemini) through classification, three-stage deduplication, auto-fix, thread replies and resolution, convergence monitoring and escalation. **Now part of the team's daily workflow**: **10** engineers, **300+** PRs processed, review time **cut in half** and false positives down **70%**
- VM Health Center** (TypeScript/Node.js + Slack Bolt + Elasticsearch, built solo from scratch) — a team dashboard and service health monitor: aggregates Jira/GitHub/Confluence into daily/weekly reports, surfaces checkout-funnel and backend health alerts, with preliminary AI root-cause analysis. **Used daily by 20+ people across 7 teams** (Product, Web, App, QA, Order, B2C), firing alerts every day and cutting incident triage time by **40%**
- Claude API in production:** SEO slug generation across **8** locales covering nearly **3,000** product pages and **~50k** API calls, with hash-based change detection skipping unchanged content to hold cost down; a Git workflow and AI code-review automation system deployable as CLI, webhook or GitHub Actions

Core Skills

AI Engineering	Claude API / Copilot SDK application development, multi-agent orchestration and convergence, AI-assisted engineering process design
Languages & Frameworks	PHP (expert, 8+ yrs), TypeScript / Node.js, Go, Python (reading proficiency); Laravel / Hyperf, Gin / go-zero, Vue
Data & Messaging	MySQL, PostgreSQL, Redis, Elasticsearch, Google BigQuery, RabbitMQ, Kafka
Cloud & DevOps	Docker, Kubernetes, AWS Lambda / GCP Cloud Run, GitHub Actions, Prometheus / Grafana / Kibana
Architecture & Domain	Microservices and BFF layering, RabbitMQ event-driven pipelines, Redis circuit breakers with two-tier kill switches, multi-tier caching and cache-key governance; commission settlement, order transaction flows, dynamic pricing

2022.11 – Present
Shanghai, China
(UTC+8)

KKday | Senior Backend Engineer

A leading global travel-experience platform operating in 50+ countries. Progressed from affiliate platform owner → B2C core backend → lead of the Pre-Purchase Conversion Task Force.

Pre-Purchase Conversion Task Force · Lead (2026 H2 – Present)

- Formed and lead a **6-person cross-platform team** (iOS/Android/Web/Backend) against a single north-star metric — conversion **1.2% → 1.8% (+50%)**; established standups, 1-on-1s and OKR alignment across the pre-purchase funnel (product page → cart → checkout)

B2C Core Transaction Flow (2025.02 – Present)

- USJ (Universal Studios Japan) booking-date changes**: re-architected across four services and authored **7** SA/SD documents. Turned scattered coupon and loyalty-point revalidation rules into an auditable decision table under an "entitlements inherit, conditions re-evaluate" principle — approved by all four teams in a single review, securing renewal with a top-tier ticketing supplier
- Major outage response (incident lead)**: traced the root cause to a microservice dependency exhausting the PHP-FPM worker pool and shipped P0+P1 hardening in **7** days (timeouts, fallbacks, a Redis circuit breaker, two-tier kill switch) — cart/validate p99 **25s+ → <2s**, iOS checkout white-screen eliminated, 5xx held below **0.01%** since

Affiliate Marketing Platform · Platform Owner (2022.11 – 2025.03)

- Top contributor and owner of two core services — affiliate-service (**56%** of all commits) and affiliate-api — as Code Owner and PR gatekeeper; integrated Google's Things To Do channel and **10+** partner channels including ShopBack, Asia Miles and Naver Shopping, growing the affiliate business **200%** YoY
- Designed the **commission settlement system**: rules engine, monthly settlement with re-run capability, BigQuery attribution, and a RabbitMQ event-driven pipeline connecting order and product data

2021.03 – 2022.10
Shanghai, China
(UTC+8)

JINGdigital | Senior Backend Engineer

A B2B MarTech/SCRM marketing-automation SaaS vendor. Responsible for backend development across a marketing-automation microservice suite (Laravel/Yii2/YAF, 10+ services).

- Owned the **JavaScript event-tracking service** (jstracking): omnichannel behavioral collection and content-marketing attribution, the primary data source for marketing automation

2017.12 – 2021.03
Shanghai, China
(UTC+8)

Shopex | Technical Expert / Lead Developer

A leading domestic e-commerce SaaS provider. Led delivery of the Apple China reseller platform and high-concurrency performance work.

- Delivered the **Apple China reseller platform** (surpassing **¥10B** GMV within 5 months of launch): integrated **15+** payment and logistics systems including SF Express and Baozun OMS, reached **98%** OMS/WMS automation, and served **3,500+** retail stores and **250K+** daily active users
- Received the company's **Golden Tomato Award** for a high-performance coupon system; raised H5 API throughput **17.5x** (**200 → 3,500 TPS**) and reached **12K TPS** under load testing (NGINX caching + TSung tuning)
- Built an SMS gateway on go-zero handling **1M+** messages/day; optimized data export from OOM failures to **10M rows in 10 minutes**

2013.06 – 2017.12
Shanghai, China
(UTC+8)

Yuanfeng Technology Group | Full-Stack Engineer

- Built a multi-merchant SaaS e-commerce platform serving **10,000+** merchants with **10-language** support, using a hybrid Google/Youdao translation pipeline that cut translation costs **70%**

Open Source & Personal Projects

- SwaggerNotes** (PHP/Laravel package, built solo): generates Swagger annotations and OpenAPI specification files; published on Packagist with **2.6k+** downloads
- knuckleswtf/scribe** (**2.3k** 🌟 Laravel API documentation generator): **2** pull requests merged upstream, listed among official contributors